

TEST PLAN

Selenium GUI Elements Automation

URL Under Test: <https://testautomationpractice.blogspot.com>

Field	Details
Document Version	1.0
Date	June 2026
Project	GUI Elements Automation
Tool	Selenium WebDriver 4.44.0
Framework	TestNG + Page Object Model
Author	Automation Engineer

1. Introduction

This Test Plan describes the strategy, scope, and approach for automating the GUI elements on the test automation practice website. The purpose of this document is to define what will be tested, how it will be tested, and what constitutes a pass or fail for each test scenario.

2. Test Objectives

The primary objectives of this test effort are:

- Verify that all GUI elements on the practice page function correctly
- Automate interaction with text fields, radio buttons, checkboxes, and dropdowns
- Validate date picker, file upload, slider, and alert handling
- Test mouse hover, double-click, and drag-and-drop user interactions
- Generate an automated HTML report with screenshots after every run
- Ensure regression coverage by maintaining a reusable test suite

3. Scope

3.1 In Scope

- All GUI elements on <https://testautomationpractice.blogspot.com>
- Text input fields: Name, Email, Phone, Address
- Radio buttons: Gender (Male / Female)
- Checkboxes: Days of the week (Monday through Sunday)
- Dropdowns: Country, Colors
- Date picker field
- File upload input
- Range slider
- Alert and Confirm popup dialogs
- Mouse Hover interaction
- Double Click action
- Drag and Drop action

3.2 Out of Scope

- Performance and load testing
- Mobile browser testing
- Backend / API testing
- Cross-browser testing beyond Chrome and Firefox

4. Test Approach

The automation framework follows the Page Object Model (POM) design pattern, separating element locators and interactions from the test logic. Selenium WebDriver 4.18.1 with

WebDriverManager is used to manage browser drivers automatically without manual configuration.

Each test case is an independent @Test method in TestNG. Tests are executed in priority order as defined in testng.xml. ExtentReports generates a dark-themed HTML report with before/after screenshots embedded for every test step.

5. Tools and Technologies

Tool / Library	Version	Purpose
Java	JDK 11+	Programming language
Selenium WebDriver	4.44.0	Browser automation
TestNG	7.9.0	Test framework and assertions
WebDriverManager	5.7.0	Automatic ChromeDriver management
ExtentReports	5.1.1	HTML test report generation
Log4j2	2.22.1	Logging throughout test execution
Maven	3.8+	Build tool and dependency management
Eclipse IDE	Latest	Development environment
Git + GitHub	Latest	Version control and code repository

6. Test Environment

Parameter	Details
Application URL	https://testautomationpractice.blogspot.com
Operating System	Windows 10/11 or macOS / Ubuntu
Browser	Google Chrome (latest), Firefox
Java Version	JDK 11 or higher
Network	Internet connection required (public site)
Execution Mode	Local execution (headed or headless)

7. Test Strategy

7.1 Test Design

Each GUI element is covered by at least one dedicated test case. Test cases are designed to:

- Interact with the element using the correct Selenium API (sendKeys, click, Select, Actions)

- Assert the expected result using TestNG Assert methods
- Capture a screenshot at key interaction points

7.2 Pass / Fail Criteria

A test case is considered PASSED when:

- All assertions pass without throwing AssertionError
- No unexpected exceptions or timeouts occur during execution
- The expected UI state is confirmed after each action

A test case is considered FAILED when:

- Any assertion fails (actual != expected)
- An element is not found within the configured explicit wait timeout
- An unexpected alert, error page, or exception occurs

8. Entry and Exit Criteria

8.1 Entry Criteria

- Test automation framework is set up and all Maven dependencies are downloaded
- Application URL is accessible from the test machine
- Google Chrome browser is installed
- All test data is configured in config.properties

8.2 Exit Criteria

- All 16 test cases have been executed at least once
- Test execution report (ExtentReport.html) is generated
- All FAILED tests have a corresponding bug report entry
- Pass rate is 90% or above

9. Risks and Mitigation

Risk	Likelihood	Mitigation
Website structure changes (locators break)	Medium	Centralize locators in Page Object, update when needed
Chrome version mismatch with ChromeDriver	Low	WebDriverManager auto-downloads matching driver
Internet unavailability during test run	Low	Run during stable network hours; retry failed tests
Flaky tests due to page load timing	Medium	Use explicit waits; avoid Thread.sleep

10. Deliverables

- Test Plan document (this document)
- Test Cases document (Excel)
- Automation source code in GitHub repository
- ExtentReport HTML execution report
- Bug Report document
- Framework Explanation document

11. Schedule

Activity	Start Date	End Date
Framework Setup and Configuration	Day 1	Day 1
Page Object Model Development	Day 1	Day 2
Test Case Implementation (TC_001–TC_016)	Day 2	Day 3
Test Execution and Debugging	Day 3	Day 4
Report Generation and Documentation	Day 4	Day 5
GitHub Push and Final Review	Day 5	Day 5